JOURNAL OF SPORTS SCIENCES
2020, VOL. 38, NO. 16, 1897–1912
https://doi.org/10.1080/02640414.2020.1763077

Routledge
Taylor & Francis Group

SPORTS PERFORMANCE

Check for updates

Overreaching and overtraining in strength sports and resistance training: A scoping review

Lee Bell, Alan Ruddock, Thomas Maden-Wilkinson and David. Rogerson

Sport and Physical Activity Research Centre, College of Health, Wellbeing and Life Sciences, Sheffield Hallam University, Sheffield, UK

ABSTRACT

To date, little is known about overreaching (OR) and the overtraining syndrome (OTS) in strength sports and resistance training (RT) populations. However, the available literature may elucidate the occurrence of both conditions in these populations. A scoping review was conducted. SPORTDiscus, Scopus and Web of Science were searched in a robust and systematic manner, with relevant articles analysed. 1170 records were retrieved during an initial search, with a total of 47 included in the review. Two broad themes were identified during data extraction: 1) overreaching in strength sports; 2) overreaching and overtraining syndrome in RT. Short-term periods of OR achieved with either high-volume or high-intensity RT can elicit functional OR (FOR) but there is also evidence that chronic high-volume and/or intensity RT can lead to non-functional overreaching (NFOR). There is minimal evidence to suggest that true OTS has occurred in strength sports or RT based on the studies entered during this review. More research is needed to develop robust guiding principles for practitioners. Additionally, due to the heterogeneous nature of the existing literature, future research would benefit from the development of practical tools to identify and diagnose the transition from FOR to NFOR, and subsequently OTS in strength athletes and RT populations.

Abbreviations: RT: Resistance training; OR: Overreaching; FOR: Functional overreaching; NFOR: Non-functional overreaching; OTS: Overtraining syndrome; WP: Weightlifting performance

ARTICLE HISTORY
Accepted 20 April 2020

KEYWORDS
Overreaching; overtraining; overtraining syndrome; strength sports; resistance training

1. Introduction

Resistance training (RT) can induce acute fatigue due to reduction in neuromuscular activation and sequencing (Raastad & Hallén, 2000), manifested as short-term (seconds to hours) decrease in performance, occurring due to impairment of central and/or peripheral mechanisms. (Todd et al., 2003) Recovery from RT varies based on the magnitude, duration and mode of training (Triscott et al., 2008), with typical mechanical and biochemical restoration occurring within 3 hours of moderate-intensity RT using loads at 70% of 1-RM or 24–96 hours during intense RT where muscular failure is achieved and/or muscle damage has occurred (Morán-Navarro et al., 2017; Raastad & Hallén, 2000; Soares et al., 2015).

Whilst a single overloading RT bout results in acute fatigue and a relative decrease in performance, short-term periods of accumulated training above the habitual level followed by subsequent recovery can lead to a super-compensation "rebound" effect known as functional overreaching (FOR), or a diminished adaptive response and long-term performance decrement, known as non-functional overreaching (NFOR). Prolonged exposure to such training may lead to overtraining syndrome (OTS) (Meeusen et al., 2013). Large-scale reviews and joint expert statements (Cadegiani & Kater, 2017; Halson & Jeukendrup, 2004; Meeusen et al., 2013) have defined these terms as such:

- *Functional overreaching (FOR)*: A short-term decrease in performance lasting days to weeks with subsequent performance supercompensation after a period of recovery.
- *Non-functional overreaching (NFOR)*: Performance decrement is observed over a period of weeks to months, and while full recovery is achieved (although not always), no super-compensation effects are achieved.
- *Overtraining syndrome (OTS)*: Long-term reductions in performance capacity observed over a period of several months.

In strengthsports such as weightlifting, it is common for athletes and coaches to utilize periods of purposeful increased volume and/or intensity RT within a competition cycle to achieve FOR (Pistilli et al., 2008; Stone et al., 2006). This may also be the case in other RT populations such as bodybuilding and high-intensity conditioning (e.g., CrossFit) where athletes participate in repeated high volume/intensity RT sessions/blocks in order to achieve a supercompensation effect (Szewczyk et al., 2018). To date, there is a lack of research into OR practices and the implications for non-functional outcomes (Kreher, 2016; Meeusen et al., 2013). As such, athletes may be at risk of maladaptation and performance decrement due to NFOR and/or OTS.

Objectives

Scoping reviews are an ideal "reconnaissance" tool to evaluate a body of literature not comprehensively reviewed, or that

CONTACT Lee Bell lee.bell@shu.ac.uk Sport and Physical Activity Research Centre, College of Health, Wellbeing and Life Sciences, Sheffield Hallam University, Sheffield, UK

© 2020 Informa UK Limited, trading as Taylor & Francis Group

exhibits a heterogeneous nature not amenable to a more systematic approach (Peters et al., 2015). Scoping reviews provide the flexibility to map out broad narratives within a limited literature base, allowing researchers to examine emerging evidence where precise lines of questioning are undetermined (Arksey and O'Malley, 2005; Munn et al., 2018). Additionally, such reviews help develop key definitions and conceptual boundaries as well as identify gaps for future research (Arksey and O'Malley, 2005).

The available OR/OTS literature in strength sports and RT contexts appears to be broad, disparate and heterogeneous. After careful investigation and consultation with the appropriate guidance (Munn et al., 2018), a scoping review was chosen ahead of a systematic review. Not only is a scoping review a relevant tool to synthesize the cross-disciplinary landscape of available evidence, it also offers a robust but transparent approach, navigating multiple-nuanced themes and identifying areas for further exploration where a systematic review is inappropriate.

A PCC framework (population, concept, context)(Peters et al., 2017) was used to develop the research question: *"what is known about overreaching and overtraining in strength sports and resistance training populations?"*

2. Methods

2.1. Protocol and registration

This scoping review was developed using guidance from the Joanna Briggs Institute (Peters et al., 2015) and PRISMA-ScR (Tricco et al., 2018), together with the methodological framework proposed by Arksey and O'Malley (Arksey and O'Malley, 2005) and subsequent recommendations from Levac et al. (Levac et al., 2010). The protocol was registered with Open Science Framework on 5 July 2019 [https://osf.io/vhp68/].

2.2. Eligibility criteria and definitions

A *strength sport* is defined here as a competitive sport where RT provides the primary overloading stimulus and where maximal strength or high force output is a primary determinant of performance, with limited or no additional overloading from endurance training. Weightlifting, powerlifting, strongman, explosive throwing sports and sprinting met these guidelines.

The term *resistance training* refers to any study where participants were required to complete a short or long-term RT-based OR protocol without additional, concurrent endurance or technical training. This included studies where both recreational RT participants and/or competitive athletes were used, or review papers that discussed RT in the context of OR/OTS.

Research was not limited by geographical location or year and was incorporated in the review if it included: 1) human participants of either sex and at all age groups; 2) OR/OTS in the context of either strength sports or RT, and; 3) peer-reviewed data including quantitative/qualitative research, prospective cohort studies, mixed-methods, systematic/scoping/narrative reviews/meta-analyses or case reports. Conference proceedings and poster presentations were not included due to potential limitations in reporting quality and/or duplication.

2.3. Information sources and search strategy

The search was conducted using three electronic databases: SPORTDiscus, Scopus and Web of Science during June 2019. These databases were selected to provide relevant literature and were identified by the research team in consultation with an expert information scientist, who assisted with the development of the search strategy and database search.

Boolean search terms AND/OR were used to identify relevant studies: "*overreaching*" OR "*overtraining*" AND *"resistance training"* OR *"strength training" OR* "*weight training"* OR "*weightlifting" OR "weight lifting"* OR *"powerlifting"* in all databases.

Titles, abstracts, keywords and data sources were searched, with relevant articles entered for full paper review. Upon completion, an investigation of additional citations from each reference list was conducted.

2.4. Data charting and synthesis

Data charting was carried out by the principal investigator using a charting tool designed for this study, developed to capture key information. The charting tool was independently assessed by the research team to determine robustness. Participant characteristics (trained/untrained, elite/non-elite, age, gender, number), and article characteristics (author, publication date, OR/OT protocols, measures and outcomes) were extracted during data charting. For comprehensive study breakdown, see Tables 1 and 2.

3. Results

3.1. Selection of sources of evidence

Initial search yielded 1170 results. After duplicates were removed, 832 studies were entered for title, abstract, keyword and data source review. Subsequently, 137 items were selected for full-text review, with 90 full texts excluded for the following reasons: 7 were symposia or poster presentations and 83 were not relevant to the central review question. Forty-seven studies were included in this scoping review (see Figure 1).

There were two broad investment themes extracted during data charting. These were developed *a priori* and agreed on by the research team:

(A) Overreaching in strength sports (Theme A) (27.7% total)
(B) Overreaching and overtraining in resistance training (Theme B) (72.3% total)

3.2. Synthesis of results

Publication dates ranged from 1987 to 2019. 20 studies (42.6%) were published between 2010 and 2019. From those 20 studies, 16 (80.0%) are located in theme B. Unlike many areas of sport science research, no obvious global increase in publications was observed, especially for theme A, further illustrating the need for future research in this area. Figure 2 highlights publications by date and theme.

Overall, 85.1% (n = 40) studies included in the review were observation/intervention studies with timelines ranging from

Table 1. Summary of data extraction for theme A: overreaching in strength sports.

Author/publication date/location	Participants	Study design	Purpose	Protocol	Measures	Study findings
(Bazyler et al., 2017), *USA*	n = 6 National Collegiate Athletic Association (NCAA) Division I track and field athletes (M: n = 4; F: n = 2; age = 20.6 ± 0.93 years)	Prospective cohort study	OR and taper on measures of muscle architecture, jumping, and throwing performance in collegiate throwers	12-wk block-periodization model consisting of throwing training and RT culminating in a 1-wk OR followed by a 3-wk taper	Throwing performance, performance measures, muscle architecture	↑ throwing performance and unloaded jump performance after OR. ↑ muscle thickness in vastus lateralis observed during in-season training but not after OR. ↓ in RT volume load multiplied by bar displacement. ↓ RPETL between in-season and OR
(C Bazyler et al., 2017), *USA*	n = 1 elite F weightlifter (age = 21.82 yrs; body mass = 70.7 kg)	Case study	Physiological and performance changes of a national-level 69 kg female weightlifter after 3 competition phases over a 28-wk training period	Observation of physiological adaptations over a 28-day period, consisting of 3 competition phases	WP, serum biomarkers, mCSA	Biomarkers of stress, inflammation and muscle hypertrophy were sensitive to training load. Performance measures indicated preparedness during competitions 1 and 2 (regional/local), but not competition 3 (national). Rapid ↓ body mass/mCSA observed during competition 3
(Fry et al., 1993), *USA*	n = 28 elite-level M junior weightlifters (age = 17.3 ± 0.3 yrs)	Double-blind, prospective cohort study	1-wk high volume weightlifting and amino acid supplementation	High-volume weightlifting exercise	WP, performance measures, serum hormones	↑ WP. ↓ vertical jump height and T concentrations in both groups. ↓ in T/C ratio was observed in placebo only
(Fry et al., 1994a), *USA*	n = 9 elite M junior weightlifters (age = 17.6 ± 0.3 yrs)	Longitudinal observational study	Effects of 1-wk of ↑ training volume on strength and serum hormones after 1-year of weightlifting	Performance testing administered before and after 1-wk OR stimulus performed at year intervals.	Performance measures, serum hormones	↑ strength by Year 2 but during either week of OR. ↓ T in year 2, but ↑ in year 2. ↑ C in year 1 and 2. T/C ratio
(Fry et al., 2000a), *USA*	n = 22 junior weightlifters (non-elite: n = 14; age = 17.2 ± 0.4 yrs; elite: n = 8; age = 18.4 ± 0.4 yrs: sex undisclosed)	Prospective cohort study	The relationship between serum testosterone and cortisol on weightlifting performance	Identical 4-wk protocol performed by participants in each group consisting of 1) 1-wk high-volume training of 3–4 training sessions/day; 2) 3-wk normal volume phase of 1–2 training sessions/day	WP, serum hormones	↑ WP by the end of wk 4. Non-elite group pre-exercise T/C alterations negatively correlated to weightlifting performance during high-volume training, with positive correlations observed during reduced volume training. between T/C and performance during high-volume training in the elite group and a positive relationship during normal volume training
(Haff et al., 2008), *USA*	n = 6 elite F weightlifters (age = 21.5 ± 3.1 yrs)	Observational study	11-wk training period performed by female weightlifters	11-wk protocol consisting of weightlifting-specific and supplemental exercises of varying volume and intensity	Performance measures, serum hormones	↑ volume load corresponded to ↓ in T/C ratio as well as concomitant ↓ in peak force during isometric peak force, dynamic pull with 30% isometric peak force and dynamic 100 kg pull trials
(Häkkinen et al., 1987), *Finland*	n = 11 elite M weightlifters (age = 23.0 ± 2.8 yrs)	Longitudinal observational study	Training volume during prolonged training on physical performance capacity and serum hormone concentrations	Weightlifting-specific training for a mean average of 5.0 (±0.8) times/wk (1.7 ± 0.4 hrs per training session, 5,200 ± 1,500 kg training volume, 80.5% ± 2.5% intensity) over the course of a competitive season (1-year)	WP, serum hormones	↓ T, T/C and T/SHBG ratio. ↑ LH during 2-wk OR period that did not return to baseline until 2-wk period of normal training and subsequent 2-wk taper had been completed. Alterations in T/SHBG correlated with weightlifting performance in the clean and jerk. Subsequent ↓ in C and LH, as well as ↑ T/SHBG during normal training/taper

(*Continued*)

Table 1. (Continued).

Author/publication date/location	Participants	Study design	Purpose	Protocol	Measures	Study findings
(Häkkinen et al., 1989), *Finland*	n = 8 elite M weightlifters (age = 24.3 ± 1.5 yrs)	Longitudinal observational study	Endocrine responses and on physical performance capacity	Weightlifting-specific training ranging from 4–10 sessions/wk (mean = 7) over the course of a competitive season (1-year)	Performance measures, serum hormones	↓ T during periods of highest training load. for serum hormones over 1 year
(Hartman et al., 2007), *USA*	n = 10 elite M weightlifters (once-daily RT: n = 5; age = 20.2 ± 1.4 yrs; twice-daily RT: n = 5; age 20.9 ± 0.9 yrs)	Prospective cohort study	Physiological responses to twice- and once-daily training sessions with similar training volumes	Twice- (8 sessions/wk) or once-daily (4 sessions/wk) weightlifting training sessions over a 3-wk period	Performance measures, serum hormones, mCSA	main effects between groups reported. ↑ %change observed in the twice-daily group for maximal isometric knee extension strength, T and T/C ratio
(Khlif et al., 2019), *Tunisia*	n = 16 elite weightlifters (M: n = 8; age = 19.46 ± 1.2 yrs; F: n =; age = 18.25 ± 1.2 yrs)	Prospective cohort study	↑ weightlifting training load on plasma iron status	3 intensive wks consisting of 10 training sessions/wk followed by 1 mod-intensity wk of 6 sessions/wk. Loads varied between 70–100 1-RM,	Serum biomarkers	Pre- to post-OR ↑ was observed in plasma ferritin, while a significant ↓ in transferrin was also observed. Plasma CK levels were ↑ pre-to post-OR. No statistically-significant changes to CRP were observed
(Pistilli et al., 2008), *USA*	Undisclosed	Observational study	1-wk weightlifting camp designed to promote a short-term period of OR; and to compare the training variables from a normal training wk to what is recommended during a short-term OR period and a taper wk	1-wk OR consisting of a 94% ↑ volume (compared to normal training), with a total ↑ of 55% (reps), 200% (sets) and 300% (number of training sessions/wk) followed by subsequent return to normal training	WP	↑ WP after 2–5 wks of normal training resumed
(Suarez et al., 2019), *USA*	n = 9 collegiate weightlifters (M: n = 4; age = 22.4 ± 1.6 yrs; F: n = 5; age = 20.5 ± 2.6 yrs)	Prospective cohort study	Kinetic and morphological adaptations that occur during distinct phases of a block periodized training cycle in weightlifters	4 training sessions/wk split into push-pull: 1) 4-wk strength-endurance phase (high volume and low-mod intensity); 2) 4-wk strength-power phase (mod volume at higher intensity); 3) 1-wk OR phase (high volume) followed by a 3-wk taper (low volume at mod intensity)	Performance measures, mCSA, muscle architecture	Small ↑ in mCSA and RFD observed across the block of training. Significant ↑ for mCSA observed during high-volume strength-endurance phase, with small ↓ in RFD. ↑ in RFD above pre-test baseline was reported during the higher-intensity strength-power phase despite significant ↓ in mCSA. Small to moderate ↑ in RFD reported during peak/taper (<150 ms. time band only), with mCSA
(Warren et al., 1992), *USA*	n = 28 elite junior weightlifters (17.3 ± 1.4 yrs)	Prospective cohort study	Short-term overwork (OR) on performance measures, blood lactate, and plasma ammonia concentrations	2-3 training sessions/day for 7-days. Training consisted of weightlifting-specific and testing conducted before and after OR training protocol	WP, performance measures, serum biomarkers	↓ jump height from pre- to post OR but snatch test performance. ↑ resting ammonia at post-test, as well as ↓ lactate and ammonia at 5-min post-exercise after OR protocol

Variations denoted as ↑ (increase) ↓ (decrease) or ↔ (no change). RT = resistance training; OR = overreaching; WP = Weightlifting performance; RPETL = session rating of perceived exertion training load; RM = repetition maximum; mCSA = muscle cross-sectional area; T = testosterone; C = cortisol; SHBG = sex hormone binding globulin; LH = luteinizing hormone; RFD = rate of force development

Table 2. Summary of data extraction for theme B: overreaching and overtraining syndrome in resistance training.

Author/publication date/location	Participants	Study design	Purpose	Protocol	Measures	Study findings
(Berning et al., 2007), *USA*	n = 6 resistance-trained M (age = 29 ± 5 yrs)	Prospective cohort study	Metabolic demands of pushing and pulling a 1,960 kg motor vehicle 400 m in an all-out maximal effort	1) single session pushing a motor vehicle 400 m; 2) single session pulling a motor vehicle 400 m	Performance measures	BLa response reached 131%, VO_2 reached 65% and HR reached 95% of values obtained from pre- VO_2 max test. Jump height ↓ pre- to post- in both pushing and pulling conditions
(Cadegiani & Kater, 2017), *Brazil*	N/A	Systematic review	Role of hormones in OTS/FOR/NFOR	N/A	Serum hormones and biomarkers	Basal hormones are unable to predict FOR/NFOR/OTS
(Cadegiani et al., 2019), *Brazil*	N/A	Post-hoc analysis	CrossFit in healthy athletes and OTS athletes	Data extracted from EROS-HPA, EROS-STRESS, EROS-BASAL, EROS-PROFILE	Serum hormones and biomarkers	90% of adaptive changes were lost in CrossFit athletes under OTS including ↓ T and ↑ E2. Long-term low CHO intake may act as a trigger for OTS
(Davies et al., 2016), *Australia*	N/A	Systematic review	Failure versus non-failure training on muscular strength	N/A	Performance measures	Similar ↑ in muscular strength are observed with failure and non-failure RT, however failure should be used sparingly to ↓ the risk of NFOR/OTS
(Drake et al., 2017), *USA*	n = 6 recreationally-active M (age = 25.0 ± 5.4 yrs)	Prospective within-subjects pre-post intervention	Short-term CF participation on measures of health and fitness	4-wks of CF training consisting of 5 training sessions/wk of either "traditional" or "real world" CrossFit	Performance measures, serum biomarkers, mood state	↑ CRP/CK and ↓ mood state observed alongside small ↑ in performance measures/fitness-based outcomes that may be indicative of FOR
(Fahey & Pearl, 1998), *USA*	n = 11 trained M (age = 22.8 ± 3.4 yrs)	Double-blind, cross-over design prospective cohort study	PS supplementation during 2-wks of resistance exercise-induced OT	5 x 10-RM for 13 exercises, 4 times/wk, for two 2-wk periods separated by a 3-wk recovery	Serum hormones	Muscle soreness ↑ in placebo group compared to PS. C concentrations were ↓ in PS compared to placebo
(Fatouros et al., 2006), *Greece*	17 resistance-trained M (age = 21.6 ± 2.6 yrs)	Prospective cohort study	cf-DNA as a potential indicator of OTS during resistance training	12-wk RT programme consisting of four 3-wk protocols: 1 and 4) low-volume exercise training 2-days/wk, 2 × 10–12 reps, 70% 1-RM; 2) high-volume exercise training 4-days/wk, 4 × 6–10 reps, 75-85% 1-RM; 3) very high-volume exercise training 6-days/wk, 6 × 6 reps, 85-100% 1-RM.	Serum biomarkers	cf-DNA ↑ proportionate to training load and may be a reliable marker of NFOR/OTS. Concomitant ↑ in CRP/UA after high- and very-high volume were also observed, with CK increasing only after very high volume training
(Fry et al., 1994b), *USA*	n = 17 resistance-trained M (age = 22.0 ± 0.9 yrs)	Prospective cohort study	The effects of a high-intensity resistance exercise OT protocol on muscular strength decrements.	Daily 10 x 1-RM (100% 1-RM) squat machine for 2-wk period	Performance measures, serum biomarkers	↓ 1-RM performance compared to control group, with subsequent ↓ in isokinetic and stimulated isometric muscle force indices. Concomitant ↑ in CK with declining performance indicative of OR
(Fry et al., 1994c), *USA*	n = 9 recreationally-trained M (age = 22.9 ± 1.3 yrs)	Prospective cohort study	Efficacy of a 3-wk, high-intensity resistance OT exercise protocol	3 weeks of 5 days/wk low-volume, high-intensity machine squats consisting of 8 single reps using 95% 1-RM with 2-min recovery between attempts. If a rep was unsuccessful, 5% of 1-RM ↓ on the subsequent attempt.	Performance measures	↓ in sprint and vertical jump performance as well as leg extension torque were observed, but 1-RM ↑. Based on ↑ strength, NFOR/OTS was not induced but results suggest FOR did occur
(Fry et al., 1994d), *USA*	n = 17 resistance-trained M (age = 22.0 ± 0.9 yrs)	Prospective cohort study	Catecholamine response to high-intensity RT OT	Daily 10 x 1-RM (100% 1-RM) squat machine for 2-wk period	Performance measures, serum biomarkers	↑ exercise-induced catecholamine concentrations, as well as ↓ responsivity of skeletal muscle to sympathetic nervous system activity reported during OT but not in control group

(*Continued*)

Table 2. (Continued).

Author/publication date/location	Participants	Study design	Purpose	Protocol	Measures	Study findings
(Fry & Kraemer, 1997), *USA*	N/A	Systematic review	Neuroendocrine responses to RT OT and OR	N/A	Serum hormones and biomarkers	RT can provide differential responses to OR/OTS during chronic exposure. Hormonal disturbances during high-volume RT may follow a pattern similar to endurance training OR/OTS, whereas during high-intensity RT, OR/OTS may result in an alternate pattern of response
(Fry et al., 1998), *USA*	n = 11 resistance-trained M (age = 22.0 ± 0.9 yrs)	Prospective cohort study	Pituitary-adrenal-gonadal responses to high-intensity RT OTS	Daily 10 x 1-RM (100% 1-RM) squat machine for 2-wk period	Performance measures, serum hormones and biomarkers	↓ in strength observed after 2-wk protocol with slight ↑ in exercise-induced T, T/C ratio as well as ↓ C. GH and plasma peptide F were unchanged.
(Fry et al., 2000b), *USA*	n = 6 resistance-trained M (age = 27.5 ± 5.4 yrs)	Prospective cohort study	3-wk high relative intensity RT resulted in OTS and to investigate what types of performance would be affected	2-days/wk for 4-wks of normal training, followed by 3-wk high-intensity RT consisting of 3 training sessions/wk of 2 × 1 95% 1-RM, 3 × 1 90% 1-RM, 3 × 10 RM	Performance measures	↑ 1-RM observed during normal training but not during high-intensity training. No change in self-reported indices of muscle soreness, vertical jump and other performance measures, but high-intensity training led to ↑ 9.1 m sprint times but ↓ peak isokinetic squat force suggesting high-intensity training may negatively impact performance measures but augment others
(Fry et al., 2001), *USA*	n = 1 resistance-trained M (age = 21 yrs)	Case study	Joint-centred mechanism of performance decrement caused by OTS	Daily 10 x 1-RM (100% 1-RM) squat machine for 2-wk period	Performance measures	↓ 1-RM and isokinetic knee-extension strength. ↑ voluntary isometric knee-extension and stimulated isometric knee-extension strength. Joint-centred overload injury of the knees as assessed by anterior/posterior drawer and Lachman tests
(Fry et al., 2006), *USA*	n = 16 resistance-trained M (age = 20.2 ± 0.1 yrs)	Prospective cohort study	Cellular and molecular responses of skeletal muscle to performance decrements due to high-relative-intensity RT OT	Daily 10 x 1-RM (100% 1-RM) squat machine for 2-wk period	Performance measures, serum biomarkers	↓ 1-RM/mean power at 100% 1-RM. ↓ in muscle β2AR density, ↑ (but insignificant) nocturnal urinary epinephrine was also observed in OTS group, as well as ↓ β2AR sensitivity. Normal training could only be resumed 2–8 wks after training cessation and may have signalled NFOR/OTS.
(Jakubowski et al., 2019), *Canada*	n = 26 trained M (age = 23 ± 2 yrs)	Randomized, double-blind repeated-measures design	HMB-FA versus leucine, added to whey protein and muscle hypertrophy and strength gains	Phase 1 consisted of 3 training sessions/wk undulating periodization for an 8-wk period, followed by phase 2; a 2-wk OR phase consisting of 5 training sessions/wk and phase 3; a subsequent 2-wk taper consisting of 3 training sessions/wk	Performance measures, serum biomarkers, mCSA	between groups for measures of mCSA or 1-RM at any phase, but an ↑ in 1-RM during the taper for both groups was indicative of FOR.
(Kraemer & Ratamess, 2005), *USA*	N/A	Literature review	Hormonal responses and adaptations to RT	N/A	Serum hormones	OR may not result in T or C alterations, and any observed change may be specific to exercise volume and/or intensity. During OT, high-intensity exercise training does not appear to alter resting T/C concentrations, whereas high-volume exercise training may result in adverse changes
(Kraemer et al., 2006, *USA)*	n = 17 trained M (age = 19.7 ± 1.4 yrs)	Double-blind, placebo-controlled, randomized study	AA supplementation on muscular performance and resting hormone concentrations during resistance training OR	4-wk OR training consisting of 4 pw resistance training. The first 2-wk were higher-volume (3 x 8-12-RM) and the last 2-wk were high-intensity (5 x 3-5-RM)	Performance measures, serum hormones and biomarkers	Significant elevation in CK was highly correlated to a ↓ in 1-RM strength during OR. ↑ SHBG was observed in the placebo group but not in the AA group, likewise, ↓ in T were observed in placebo but not in AA

(*Continued*)

Table 2. (Continued).

Author/publication date/location	Participants	Study design	Purpose	Protocol	Measures	Study findings
(Lowery et al., 2016), *USA*	n = 17 trained men (age = 21.7 ± 0.4 yrs)	Double-blind, placebo-controlled, randomized study	12-wk HMB-FA and ATP supplementation on lean body mass, strength and power	12-wk protocol consisting of three phases: 1) 8-wk daily undulating periodization; 2) 2-wk OR; 3) 2-wk taper	Performance measures, serum hormones and biomarkers, mCSA	During OR, both strength and power ↓ in the placebo group compared to a small ↑ in the HMB-FA/ATP group. LBM, strength and performance improved similarly in both groups after taper
(Nicoll et al., 2016), *USA*	n = 14 resistance-trained M (age = 19.8 ± 1.8 yrs)	Prospective cohort study	ERK1/2, JNK and p38-MAPK following a period of RT OR/OT	HPOR consisting of 2-wk normal training followed by a 2-wk OR phase of twice-daily workouts for 7.5 days (15 total workouts), or HIOR protocol consisting of 4-wk normal training followed by daily 10 x 1-RM (100% 1-RM) squat machine for 2-wk period	Serum biomarkers	Total-ERK ↑ after HPOR but ↓ with HIOT. Additionally, the HIOT group also experienced a ↓ in p-p38-MAPK, and the HPOR group reported a ↓ in the ratio of p-JNK/total-JNK and p-ERK1/2/total-ERK, suggesting that β2AR expression may be altered during periods of OR/OT
(Raastad et al., 2001), *Norway*	n = 18 resistance-trained M (heavy training group: age = 25.9 ± 1.3 yrs; control: age = 25.4 ± 1.3 yrs)	Prospective cohort study	Neuromuscular and hormone changes during 2 wks of heavy strength training	2-wk heavy lower body training consisting of 6-days/wk leg press/hack squat and squat/front squat on the 7th. All exercises were 3 x 6-RM	Performance measures, serum hormones and biomarkers	↓ T coincided with ↑ maximal strength. ↑ in urinary catecholamine levels during heavy training was also reported. in control group
(Raeder et al., 2016), *Germany*	n = 23 athletes (M: n = 14; age = 24.1 ± 2.0 yrs.; F: n = 9; age = 25.4 ± 1.9)	Prospective cohort study	Neuromuscular, physiological, and perceptual markers for routine assessment of fatigue and recovery in high-intensity RT	6-day intensified strength training microcycle consisting of twice-daily RT	Performance measures, muscle contractile properties, serum biomarkers, psychometric measures	Maximal strength ↓ during intensified training, however, performance returned to baseline after 3-days of recovery. Other performance measures remained ↓ beyond this point. Measures of stress-recovery, muscle damage, soreness and neuromuscular function may have detected the onset of NFOR
(Ratamess et al., 2003), *USA*	n = 17 trained M (AA: n = 9; age = 19.7 ± 1.4 yrs; placebo: n = 8; age = 21.3 ± 3.0 yrs)	Double-blind, placebo-controlled, randomized study	AA supplementation on muscular strength, power, and high-intensity endurance during short-term RT OR	4-wk OR consisting of 2-wk mod-intensity/high-volume (3 x 8-12-RM, 8 exercises) and 2-wk high-int, low volume (5 x 3-5-RM, 5 exercises), followed by 2-wk taper	Performance measures	Initial ↓ 1-RM strength and peak power was observed during OR in the placebo group only, however, ↑ 1-RM strength in both groups was observed after taper
(Robbins et al., 2012), *Australia*	n = 43 resistance-trained M (1-set: n = 11; age = 25.5 ± 1.4 yrs; 4-set: n = 11; age = 31.0 ± 3.2 yrs; 8-set: n = 10; age = 26.0 ± 1.5 yrs)	Prospective cohort study	Lower-body strength in RT men after performing varying training volumes (1, 4, 8 sets)	2-wk standardization phase, 3-wk volume-manipulated (VM) phase, 4-wk post-manipulation phase. During VM, participants completed either 1, 4 or 8-sets of back squat at 80%-1-RM additional to other exercises.	Performance measures	All groups ↑ 1-RM at wk-6, however the magnitude of change was largest in the 8-set group, suggesting a potential FOR dose-response over short-term periods
(Sharp & Pearson, 2010), *USA*	n = 8 trained M (age = 22.9 ± 2.0 yrs)	Balanced, cross-over, placebo-controlled, double-blind, repeated measures design,	Short-term amino acid supplementation on anabolic hormonal profile and muscle cell damage during a period of high-intensity RT OR	1-wk high-intensity OR (4 training sessions/wk) consisting of 3 x 6-RM, 8 exercises)	Serum hormones and biomarkers	Serum T was significantly higher in the AA group compared to placebo, and C and CK were ↓

(*Continued*)

Table 2. (Continued).

Author/publication date/location	Participants	Study design	Purpose	Protocol	Measures	Study findings
(Sikorski et al., 2013), *USA*	n = 35 resistance-trained participants (undisclosed sex) (age = 21.3 ± 1.9 yrs)	Prospective cohort study	High-volume muscle damaging training session on perceived recovery	A single exercise session consisting of 3 x 10-12-RM involving upper and lower body RT	Perceived recovery scale (PRS), serum hormones and biomarkers	PRS ↓ and ratings of soreness/CK concentrations ↑ pre- to post-training. to T or C, however free T may have a positive relationship with PRS
(Sterczala et al., 2017), *USA*	n = 17 trained M (supplement: n = 8; age = 22.6 ± 2.1 yrs; placebo: n = 3; age = 20.3 ± 1.3 yrs; control: n = 6; age = 24.2 ± 4.6 yrs)	Randomized, double blind, group × time experimental design	Multi-ingredient recovery drink on OR induced by high-frequency, high-power RT	1-wk OR protocol consisting of 10 × 5 reps speed squats (submaximal repetitions performed at maximal speed) twice-daily for a total of 15 sessions	Performance measures, serum biomarkers	↓ mean squat velocity and power using 70%-1RM observed in both groups after OR, but 1-RM was unaffected. ↓ β2-AR expression also observed (61% in the supplement group, 81% in the placebo group)
(Stone et al., 2000), *USA*	n = 21 resistance-trained M (age not reported)	Prospective cohort study	Different weight training approaches on maximal strength over 12-wks	3 groups: 1) Non-periodized training (5 x 6-RM); 2) stepwise periodization group (↓ volume in steps); and 3) OR group (fewer reps)	Performance measures	Stepwise and OR periodization led to ↑ 1-RM compared to the non-periodized group. The OR group reported the greatest relative training intensity of 1-RM at 72%, compared to 67% in the non-periodized group and 61% for the stepwise group
(Taylor et al., 2016), *Australia*	n = 6 strength-trained athletes (M: n = 3; age = 28.0 ± 5.9 yrs; F: n = 3; age = 28.0 ± 0.7 yrs)	Prospective cohort study	Within-athlete variability exist in counter-movement jump performance	12 wk: baseline (4-wk), OR (4-wk) and taper (4-wk). During the OR phase, training volume ↑ by approximately 10% each wk	Performance measures	Within-subject variability in jump performance was observed during all phases of training, but was highest during OR
(Tibana et al., 2018), *Brazil*	n = 9 trained M (age = 26.7 ± 6.6 yrs)	Prospective cohort study	Two different extreme conditioning sessions on neuromuscular responses and biomarkers in trained men	2 training sessions (24-hrs apart) using a variety of RT exercises, intensities and volumes	Performance measures, serum biomarkers	A single bout of exercise led to ↑ BLa and cytokine (IL-6, IL-10 and OST) concentrations from workout one to two, for muscle power
(Volek et al., 2004), *USA*	n = 17 trained M (CrM: n = 9; age = 20.7 ± 1.9 yrs; placebo: n = 8; age = 21.3 ± 3.0 yrs)	Double-blind, placebo-controlled, randomized study	Creatine monohydrate (CrM) supplementation during short-term RT OR	4-wk OR consisting of 2-wk mod-intensity/high-volume (3 x 8-12-RM, 8 exercises) and 2-wk high-int, low volume (5 x 3-5-RM, 5 exercises), followed by 2-wk taper	Performance measures, serum hormones	↓ maximal and explosive power during OR in the placebo group but not CrM. A tendency for improvement in CrM following subsequent 2-wk taper was also observed. T and T/SHBG both ↓ during OR, with ↑ C in CrM at wk-1 but return to baseline at wk 2
(Willardson, 2007), *USA*	N/A	Literature review	The application of training to failure	N/A	Performance measures	Chronic exposure to failure training may result in OTS and/or overuse injury
(Willardson et al., 2010), *USA*	N/A	Literature review	Application of training to failure in mainstream RT programmes	N/A	Performance measures	Chronic exposure to failure training may result in OTS and/or overuse injury
(Wilson et al., 2013), *USA*	n = 21 trained M (age = 23.4 ± 0.7 yrs)	Double-blind, placebo-controlled, randomized study	Oral ATP supplementation on muscular adaptations in trained individuals during OR training	12-wk protocol consisting of three phases: 1) 8-wk daily undulating periodization; 2) 2-wk OR; 3) 2-wk taper	Performance measures, serum hormones and biomarkers, mCSA	ATP and placebo both ↑ maximal strength, muscle power and LBM, with ATP resulting in significant improvements when compared to placebo. During OR, both groups experienced ↓ maximal strength and muscle power, however placebo was significantly ↓ than ATP. ↑ in CK in both groups during OR were observed, but ↑ performance suggests FOR occurred in both groups

Variations denoted as ↑ (increase) ↓ (decrease) or ↔ (no change). RT = resistance training; OR = overreaching; FOR = functional overreaching; NFOR = non-functional overreaching; OTS = overtraining syndrome; CHO = carbohydrate; T = testosterone; E2 = oestradiol; C = cortisol; BLA = blood lactate; $\dot{V}O_2$ = oxygen uptake; CF = CrossFit; CK = creatine kinase; CRP = C-reactive protein; PS = phosphatidylserine; cf-DNA = cell-free deoxyribonucleic acid; UA = uric acid; RM = repetition maximum; β2AR = beta-2-adrenergic receptor; AA = amino acid; SHBG = sex hormone binding globulin; HMB = β-Hydroxy β-methylbutyric-free acid; ATP = adenosine-5′-triphosphate; ERK = extracellular signal-regulated kinase; JNK = c-Jun NH2-terminal kinase; HPOR = high-power overreaching; HIOR = high-intensity overreaching; MAPK = mitogen-activated protein kinase; PRS = perceived recovery scale; IL = interleukin; OST = osteoprotegerin = CrM = creatine monohydrate; LBM = lean body mass; mCSA = muscle cross-sectional area.

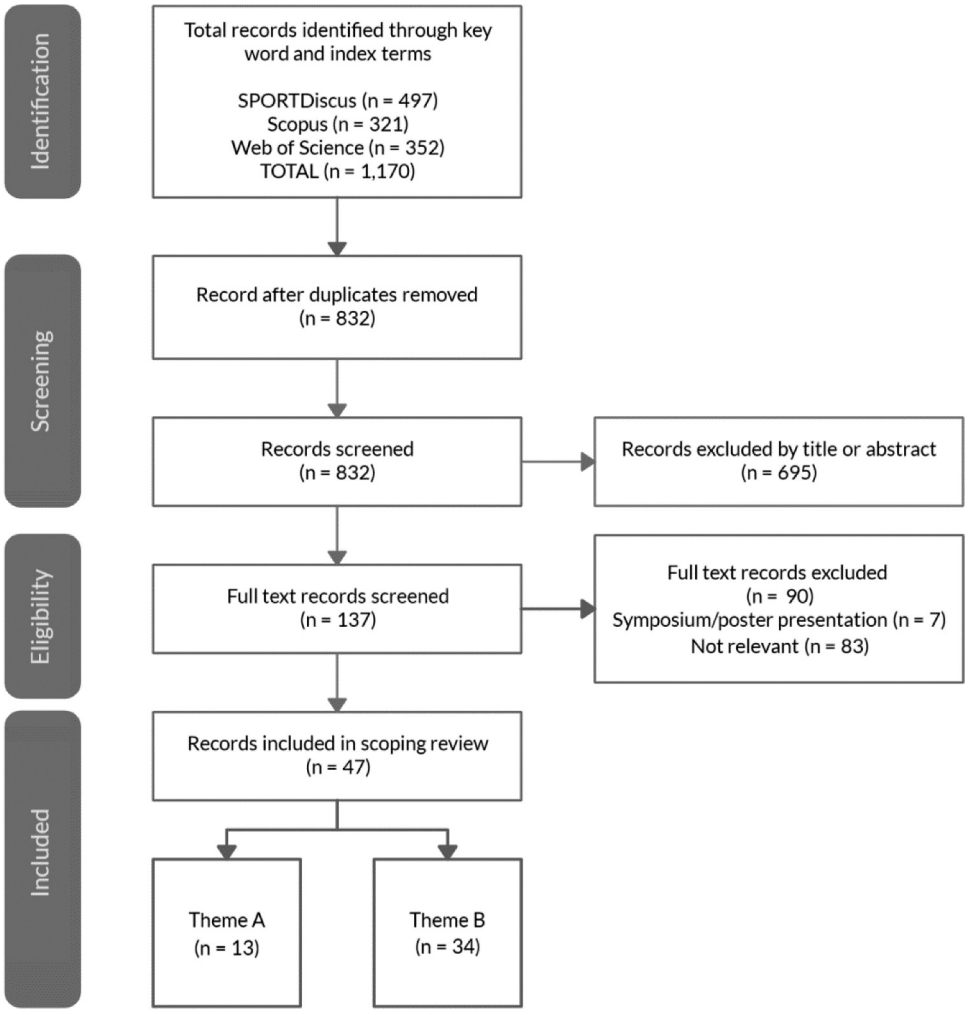

Figure 1. PRISMA-ScR *Flow chart of extracted, included and excluded studies.*

a single training session to a ~ 1-year competition schedule. Participant cohort sizes ranged from 1 to 28 in theme A and 1–43 in theme B.

2 studies used female-only participants (1 was a single subject case study) and 5 included both males and females. 4 studies did not disclose the sex of the participants. 29 used only male participants. 4 studies involved junior weightlifters.

In theme A, 92.3% (n = 12) of studies involved weightlifters, with a single study (7.7%) involving throwing athletes. No studies included powerlifting, strongman or sprinting athletes.

There were 6 systematic/literature reviews included in data extraction, with 1 post-hoc data analysis also included (all located within theme B). 23.5% of studies for Theme B (n = 8) were conducted by the same first author; Andrew C. Fry, published between 1994 and 2006. In theme A, the same author published 2 studies (15.4%) as the first author between 1993 and 1994.

Several measures and tools were used across both themes to identify the onset of OR/OTS and fell under the following broad categories: "weightlifting performance (WP)", "performance measures", "serum hormones" or "serum biomarkers" (see Tables 1 and 2). These are discussed in detail in the discussion.

Despite the ongoing risk of NFOR, our review considers this area to be underinvestigated based on a lack of studies investigating OR/OTS in strength sports and RT, as well as insufficient data relating to diagnoses and guidelines for coaches.

4. Discussion

4.1. Theme A: Overreaching in strength sports

4.1.1. Endocrinological measures and biomarkers

Studies have investigated the effects of purposeful OR on serum hormones and biomarkers with conflicting results (Bazyler et al., 2017; Fahey & Pearl, 1998; AC Fry et al., 2000a, 1994a, 1993; Haff et al., 2008; Häkkinen et al., 1989, 1987; Hartman et al., 2007; Khlif et al., 2019; Warren et al., 1992). Most studies involved elite-level weightlifters, with a single study comparing the effects of OR in elite- and non-elite participants (AC Fry et al., 2000a). 2 studies involved a female-only cohort (Bazyler et al., 2017; Haff et al., 2008) and 4 studies recruited junior-level participants (AC Fry et al., 2000a, 1994a, 1993; Warren et al., 1992). Collectively, the data indicate that endocrinological markers can detect acute changes in training state, which could be useful for monitoring purposes in

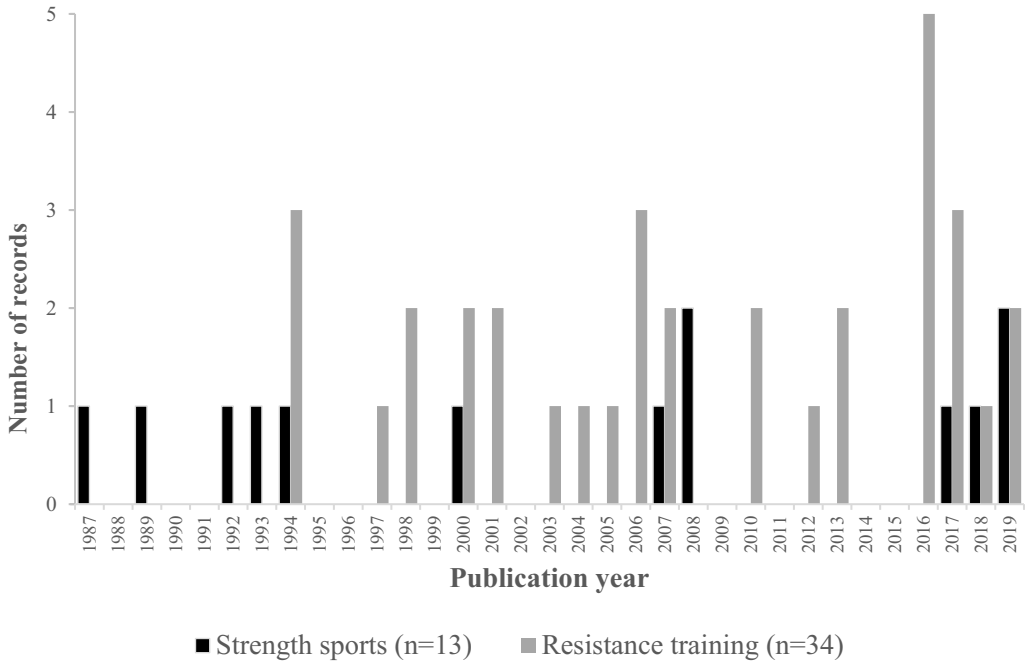

Figure 2. Publications for strength sports (Theme A, n = 13; Black) and resistance training (Theme B, n = 34; Grey) meeting search criteria for overreaching and/or overtraining.

practice. However, these studies also indicate that such hormonal alterations can occur independent of performance changes, and therefore may not be reliable when identifying NFOR.

Testosterone (T), cortisol (C) and T/C ratio were investigated in several short- and long-term studies (Bazyler et al., 2017; Fry et al., 2000a, 1994a, 1993; Haff et al., 2008; Häkkinen et al., 1989, 1987; Hartman et al., 2007), with additional measures of T and sex hormone-binding globulin (SHBG) assessed during a single study (Häkkinen et al., 1987). Overall, results indicated that training state and anabolic-catabolic response to RT were somewhat related to T/C changes (Bazyler et al., 2017; Haff et al., 2008; Häkkinen et al., 1987); however, disturbances were independent of performance in others (Fry et al., 2000a, 1993; Häkkinen et al., 1989; Hartman et al., 2007). Such effects might indicate that these biomarkers are not appropriate for the identification of NFOR.

Lifting experience was found to influence both acute (Fry et al., 2000a) and chronic (Fry et al., 1994a) hormonal responses to OR. Attenuation of T/C was observed after 1 week of purposeful OR in year 1; however, it was augmented in year 2 during a longitudinal observation of male junior weightlifters (Fry et al., 1994a). Endocrinological responses differed between elite- and non-elite weightlifters during short-term OR, with elite participants experiencing no relationship between T/C ratio and OR, however, a negative correlation between T/C and OR training in non-elite participants was observed suggesting training competence and/or experience might affect the hormonal response to overload training.

Other potential biomarkers have been assessed during purposeful OR, including lactate and ammonia (Warren et al., 1992), creatine kinase (CK) and plasma iron status (Khlif et al., 2019). Short-term OR resulted in altered lactate and ammonia concentrations coinciding with reductions in maximum effort jump performance (Warren et al., 1992). However, no decrease in performance during a weightlifting-specific snatch test was observed, suggesting alterations in such biomarkers can occur without performance decrement.

A single study assessed the effects of short-term OR on plasma iron status and selective inflammatory markers (Khlif et al., 2019). OR resulted in altered iron-status-balance parameters (especially in female weightlifters), with concomitant increases in CK. Whilst altered iron status coincided with RT-based OR, performance was not measured; therefore, the relationship between iron status and performance is unknown. However, such alterations might indicate acute adaptation rather than detection of NFOR with more research needed to investigate further.

In summary, endocrinological responses to RT may help coaches identify acute training state, trainability, and long-term adaptive responses to exercise. However, endocrinological alterations can occur independent of performance, and whilst T/C may identify acute hormonal disturbance in strength athletes, does not appear to be a determinant of FOR/NFOR. Changes in iron metabolism and selective biomarkers caused by short-term changes in RT may provide a novel method for identifying FOR/NFOR in elite-level weightlifters during intensified training protocols; current evidence, however, is limited and requires further investigation.

4.1.2. Performance measures

Several studies have assessed the effects of short-term, purposeful OR on performance outcomes in weightlifters, including weightlifting-specific testing (Bazyler et al., 2017; Fry et al., 2000a; Häkkinen et al., 1987; Pistilli et al., 2008; Warren et al., 1992)/or general measures of performance. (Fry et al., 1994a; Haff et al., 2008; Häkkinen et al., 1989; Hartman et al., 2007;

Suarez et al., 2019; Warren et al., 1992). A single study examined throwing performance parameters in Collegiate-level throwers (Bazyler et al., 2017). Overall, changes in maximal force (MF), rate of force development (RFD) and relative peak power output (PPO) through maximal effort jumping (Bazyler et al., 2017; Bazyler et al., 2017; Hartman et al., 2007; Warren et al., 1992) and isometric- and/or dynamic mid-thigh pull (IMTP/DMTP) testing (Haff et al., 2008; Suarez et al., 2019) are able to identify changes in training load and fatigue which may occur prior to performance decrement; however, no performance test can currently determine the onset of NFOR.

Periods of increased RT volume have resulted in weightlifting-specific performance improvement in elite/non-elite-level weightlifters (Fry et al., 2000a; Häkkinen et al., 1989; Pistilli et al., 2008; Suarez et al., 2019) and throwers (Bazyler et al., 2017), suggesting that FOR can be achieved through manipulation of training and can be successfully monitored using performance measures. Some studies found that high-volume RT did not lead to changes in weightlifting-specific performance but did result in reduced maximal jump height during purposeful OR (Fry et al., 1993; Warren et al., 1992), suggesting that coaches could use some performance measures to guide successful programming and avoid NFOR. Measures of RFD and MF appear to be sensitive to changes in training load and may provide a viable method to determine changes in neuromuscular characteristics prior to performance decrement, especially with tests that mirror the technical constraints of weightlifting and throwing-specific performance. Further research will elucidate whether such testing can differentiate between typical fatigue associated with increased training load indicative of FOR, or the onset of NFOR.

4.1.3 Changes in body mass and muscle cross-sectional area

Reductions in body mass (−6.0 kg) and a very likely reduction in muscle cross-sectional area of the vastus lateralis (precision = 99%, effect size = 2.08) were observed in a female athlete during an OR phase in preparation for a national weightlifting competition (Bazyler et al., 2017). Nonsignificant increases in cross-sectional area of the vastus lateralis ($p > .05$) were observed in male weightlifters training once- or twice-daily for 3-weeks, with researchers speculating that multiple daily training sessions may be more beneficial to muscle hypertrophy and may also offer a more effective neuromuscular training stimulus (Hartman et al., 2007). There is limited literature available in the area of OR and muscle hypertrophy in strength sports, and it is unwise to offer a generalised consensus based on the available literature, but an investigation into the effects of weight cutting through caloric restriction during OR is needed to expand current understanding in this area.

4.2. Theme B: Overreaching and overtraining in resistance exercise

4.2.1. The effects of high-volume and high-intensity resistance training on performance measures

Studies have examined the effects of high-volume and/or high-intensity RT (Berning et al., 2007; Fry et al., 2001, 1994c, 1998, 1994b, 1994d, 2006, 2000b; Nicoll et al., 2016; Raastad et al., 2001; Raeder et al., 2016; Robbins et al., 2012; Sharp & Pearson, 2010; Sikorski et al., 2013; Sterczala et al., 2017; Taylor et al., 2016; Tibana et al., 2018) as well as phasic, periodized approaches to OR (Fatouros et al., 2006; Jakubowski et al., 2019; Kraemer et al., 2006; Lowery et al., 2016; Ratamess et al., 2003; Stone et al., 2000; Volek et al., 2004; Wilson et al., 2013). Overall, both short-term increased RT volume and intensity can result in FOR, but excessive exposure in terms of magnitude and/or duration can also result in NFOR – especially during prolonged high-intensity RT. Identifying the transitional threshold from FOR to NFOR is difficult to determine based on the limited, heterogenous studies available within the literature and practitioners need to be aware that optimising performance is reliant on the prescription of appropriate volume and intensity during purposeful OR.

Increases in RT volume designed to induce fatigue over a 4-week period have resulted in performance improvement indicative of FOR (Taylor et al., 2016). Similarly, increases in intensity-matched lower body RT volume over a 6-week period have resulted in improved maximal strength, with higher volumes superior to moderate or low volume (Robbins et al., 2012). Reductions in mean power (MP) jumping performance have illustrated greater within-subject variability during periodized overload training, which was greatest during OR, suggesting that whilst jump performance can be used to assess temporal changes to MF, MP, RFD and fatigue in RT populations during OR, practitioners need to be aware of typical error scores when interpreting change from subject to subject (Taylor et al., 2016).

Differing definitions of "high-volume" training makes inferencing from the existing data challenging. For example, a single upper/lower body RT workout of 3 × 10–12 RM was defined as "high volume, muscle damaging training" in one study (Sikorski et al., 2013) which was notably lower in magnitude and duration (of volume) compared to other high-volume RT studies (Nicoll et al., 2016; Raeder et al., 2016; Robbins et al., 2012; Taylor et al., 2016). Such ambiguity raises questions around the thresholds used to determine high-volume OR in the RT literature, which appear to be poorly defined.

A lack of consistent findings exists within high-intensity OR literature, with short-term, purposeful high-intensity OR of similar duration resulting in FOR in some studies (Fry et al., 1994c) but performance plateau (Fry et al., 2000b) and NFOR in others (Fry et al., 1998, 1994b, 2006; Raeder et al., 2016). Although designed to invoke OTS, an increase in maximal strength was observed after a 2-week high-intensity OR protocol in 9 recreationally trained men, even in the presence of reduced jumping and sprint performance (Fry et al., 1994c) Conversely, maximal strength plateaued during a purposeful 3-week OR period (after a period of strength improvement in a preceding 4-week period of normal RT), with an associated reduction in peak isokinetic squat force at 0.20 m x s-1 (Fry et al., 2000b). However, several other performance parameters including vertical jump height, 36.6-m sprints, lateral agility and isokinetic squat force at 0.82 and 1.43 m x s-1 were unchanged.

Maximal strength decline with concomitant reductions in isokinetic and stimulated isometric muscle force has been observed during periods of high-intensity OR (Fry et al., 1994b). Similarly, short-term decreases in maximal strength have been observed after 2–3 week purposeful OR (Fry et al.,

1998), and in maximal strength/mean power at 1-RM after 2-week OR (Fry et al., 2006). A decrease in maximal strength and isokinetic knee-extension strength, as well as joint-centred overload injury of the knee, occurred in a male subject after high-intensity OR, suggesting indices of NFOR could place athletes at a higher risk of injury (Fry et al., 2001). Interestingly, voluntary isometric knee-extension and stimulated isometric knee-extension strength increased, suggesting a reduction in performance only affected dynamic strength indices.

Whilst a variety of high-intensity protocols have been observed in the literature, several studies utilised the same protocol, consisting of daily 10 x 1-RM (Fry et al., 2001, 1998, 1994b, 2006), resulting in performance decrements indicative of NFOR and/or OTS. Such an extreme protocol is unlikely to reflect RT practice, but suggests that a dose-response "threshold" might exist by demonstrating that periods of high-intensity RT can result in NFOR even over short periods of time. One study stated that after completing the 10 x 1-RM protocol, normal training could only be resumed after 2–8 weeks of cessation and may be one of the few available studies in the literature where OTS occurred, based on current definitions (Fry et al., 2006).

Overall, high-volume and high-intensity RT programming can result in FOR; however, in the presence of excessive magnitude or duration of RT can also result in NFOR. Whilst evidence might suggest a dose-response to both volume and intensity exists, currently, there is minimal literature to support at what point that might occur. Further research to determine when FOR capacity is maximized and non-functional effects begin to take place will help coaches programme optimal RT and avoid unnecessary performance decline.

4.2.2. Extreme conditioning practices

Studies have investigated the high-volume, high-intensity and multi-adaptive nature of CrossFit (CF) and other extreme RT-based conditioning practices (Cadegiani et al., 2019; Drake et al., 2017; Tibana et al., 2018). Findings suggest that such RT practices can lead to endocrinological alteration (Cadegiani et al., 2019), increased inflammatory response (Tibana et al., 2018), performance decline and increased risk of NFOR and possibly OTS if the balance between training and recovery is not adhered to (Cadegiani et al., 2019; Drake et al., 2017).

Participation in two consecutive "extreme conditioning" workouts resulted in increased anti-inflammatory cytokines without impacting muscle function (Tibana et al., 2018) suggesting that acute disturbances in some biomarkers are sensitive to short-term increases during high-intensity, high-volume RT prior to performance changes being observed. Whilst short-term CF-based RT can result in FOR, increased acute inflammatory markers and measures of mood state may suggest that prolonged exposure could lead to an NFOR state (Drake et al., 2017). In a post-hoc analysis of data, those who participated in high-intensity CF training and displayed OTS (defined as fulfiling diagnostic criteria outlined by Meeusen et al. (Meeusen et al., 2013)), reported hormonal and biochemical disturbances including attenuated T and elevated oestradiol (E2) (Cadegiani et al., 2019). Interestingly, performance decline associated with OTS may have been accelerated by insufficient carbohydrate (CHO) intakes, which were 3 times lower in the OTS CF group compared to healthy CF athletes. The author suggested that compensatory high-CHO meals may offer a protective role against OTS, and that CHO intakes of <5.0 g/kg/day for 8-weeks or more may be a contributing cause of OTS in CF athletes.

Literature in CF populations appears to be limited; however, based on existing evidence, the risk of NFOR/appears to be high in these athletes – especially in the presence of restricted CHO intake. In high-intensity, extreme RT-based exercise such as CF, a robust series of systemic markers should be developed to help coaches optimize training (Drake et al., 2017).

4.2.3. Training to muscular failure

Three review papers have discussed the potential effects of muscular failure on OR/OTS (Davies et al., 2016; Willardson, 2007; Willardson et al., 2010); however, no studies appear to have tested this hypothesis in practice. Whilst regular failure training can stimulate increases in strength and athletic performance, it might also result in NFOR/OTS when used to excess (Davies et al., 2016; Willardson, 2007; Willardson et al., 2010). Consequently, future research should investigate the effects of training to muscular failure on NFOR/OTS to inform the scientific community, (Davies et al., 2016) and to provide coaches and athletes with much-needed guidance in this area – particularly in experienced recreational bodybuilders where training to failure may be more widely used (Willardson et al., 2010).

4.2.4. Endocrinological measures and biomarkers

The effects of chronic RT exposure on hormonal and biochemical systems have been investigated in several cohort studies (Drake et al., 2017; Fatouros et al., 2006; Fry et al., 1998, 1994b, 1994d, 2006; Nicoll et al., 2016; Raastad et al., 2001; Raeder et al., 2016; Sikorski et al., 2013; Tibana et al., 2018) with two systematic reviews investigating the hormonal response to OR, (Cadegiani & Kater, 2017; Davies et al., 2016) a post-hoc analysis (Cadegiani et al., 2019) and a single review referencing OTS within the broader theme of RT (Fry & Kraemer, 1997). Overall, chronic exposure to RT can lead to altered hormonal responses, with high-volume RT following similar patterns to endurance training, whereas high-intensity RT appears to result in a differential response (Fry & Kraemer, 1997). It is unlikely that endocrinological measures can predict NFOR/OTS due to a lack of reliable supporting literature but may be useful to identify acute changes in training state. The heterogeneous nature of the literature coupled with inconsistent findings makes it difficult to corroborate the effects of purposeful OR on hormonal systems and it is likely that serum hormones such as T, C and T/C ratio are able to predict the onset of NFOR/OTS(Cadegiani & Kater, 2017; Kraemer & Ratamess, 2005; Sikorski et al., 2013).

A recent systematic review suggested basal hormone levels are not a reliable predictor of NFOR/OTS and that excessive endurance and/or RT practices could lead to neuroendocrine fluctuations indicative of NFOR (Cadegiani & Kater, 2017). From 38 studies included in this review, only 3 (7.9%) investigated RT; therefore, only limited literature relevant to this review was included. A small increase in T and T/C ratio was observed during 2-weeks of daily high-intensity RT (Fry et al., 1998), and whilst 1-RM performance decreased over the training period,

measures of anabolic status were unable to detect such strength loss. Conversely, reductions in T coincided with an increase in maximal strength during 2-weeks of heavy training (Raastad et al., 2001) and in CF participants classified as suffering OTS (Cadegiani et al., 2019), illustrating a lack of consistency in hormonal responses to high-intensity RT.

Increases in C-reactive protein (CRP) and CK have been reported during periods of short-term OR(Drake et al., 2017; Fatouros et al., 2006; Sikorski et al., 2013). CK appears to be correlated with perceived recovery after a single RT session (Sikorski et al., 2013), and maximal squat strength (Kraemer et al., 2006). However, increased CRP/CK has been reported after a period of high-intensity, high-volume RT resulting in performance gain (Drake et al., 2017), and therefore may be a part of the adaptive process that underpins FOR rather than an indicator of NFOR or OTS. A slight but insignificant increase in nocturnal urinary epinephrine activity during high-intensity RT has been observed, as well as significant downregulation in β2-adrenergic receptors concurrently with decreased 1-RM strength (Fry et al., 2006). Interestingly, OTS was diagnosed through a reduction in maximal strength as well as mean power at 100% 1-RM loads, which is conflicting with current diagnostic criteria (Meeusen et al., 2013) based on retrospective recovery time-course. That said, normal training could be resumed only after 2–8 week suggesting NFOR was more likely to have occurred in some participants, but OTS could have occurred in others.

Changes in skeletal muscle signal transduction downstream to β2-adrenergic receptors have been observed in trained males during 2–4 weeks of high-intensity and high-power OR (Nicoll et al., 2016). Similar reductions in β2-adrenergic expression and increased nocturnal urinary epinephrine have also been observed during high-intensity OR (Sterczala et al., 2017), and as such, RT participants may experience alterations in epinephrine-β2-ERK signalling axis during periods of OR.

A novel marker of cell-free plasma DNA (cf-DNA) has been proposed as a possible tool to detect OTS during intensified RT (Fatouros et al., 2006). cf-DNA increased proportionate to training load during a 12-week period of undulating RT volume, with highest concentrations reported during a high-volume, high-intensity phase. cf-DNA may be sensitive enough to detect short-term response to OR, with transient increases in cf-DNA possibly due to increased muscle fibre damage caused by prolonged overloading RT.

Hormonal and biochemical measures may help coaches identify acute training state, but inconsistent findings in the literature suggest these measures are not sufficiently robust to identify or predict NFOR/OTS; particularly when used in isolation rather than in combination with performance measures.

4.2.5. Supplement use during periods of overreaching periods

The effects of various supplements have been examined during purposeful OR periods, including amino acids (AA) (Kraemer et al., 2006; Ratamess et al., 2003; Sharp & Pearson, 2010), β-Hydroxy β-methylbutyric-free acid (HMB-FA)(Jakubowski et al., 2019), adenosine-5′-triphosphate (ATP)(Wilson et al., 2013), creatine monohydrate (CrM)(Volek et al., 2004), phosphatidylserine (Ps)(Fahey & Pearl, 1998) and multi-ingredient supplementation (Lowery et al., 2016; Sterczala et al., 2017). Overall, supplementation during purposeful OR may help to offset the deleterious effects of NFOR; however, some studies have observed no change. The overall literature to support supplementation during OR RT is less than conclusive.

High-volume OR initially leads to a decrease in muscle strength and power, with subsequent rebound effect (FOR) after low volume training in both AA supplementation and placebo (Ratamess et al., 2003). Similarly, a small decrease in 1-RM was observed during high-volume OR in both whey and HMB-FA and whey and leucine groups. However, increased maximal strength was observed during a subsequent taper, indicative of FOR (Jakubowski et al., 2019). No differences were observed at any point between groups, suggesting the OR training stimulus and not supplementation was the main driver of FOR outcomes. AA supplementation was found to preserve T and attenuate CK levels during high-volume RT and was highly correlated to reductions in squat 1-RM but a subsequent increase in maximal strength suggests the protocol resulted in FOR (Kraemer et al., 2006).

High-power RT OR led to a reduction in mean squat velocity and a concurrent increase in the ratio between serum epinephrine/β2-adrenergic receptor expression (β2-AR), without a reduction in maximal strength in both a multi-ingredient supplementation group and control (Sterczala et al., 2017). Whilst these results further illustrate the attenuation in force and velocity prior to maximal strength decay during periods of intensified RT, it is worth noting that maximal strength did begin to plateau, and further stressful RT could have resulted in NFOR. The group receiving supplementation demonstrated a smaller decrease in β2-AR expression and a lower epinephrine/β2-AR ratio, suggesting the recovery drink reduced the detrimental effects of OR on sympathetic activity, however, had no effect on performance outcome compared to placebo. Interestingly, the authors of this study referred to the 1-wk training phase as an "overtraining" phase; however, no incidence of OTS occurred, although some participants were overreached based on time-course to recovery. An exploration into differences between high-power OR and high intensity/volume OR may provide further understanding in this area.

4.2.6. Changes in body mass and muscle cross-sectional area

A small number of studies have investigated the effects of dietary supplementation on alterations to lean body mass (LBM) and mCSA during strength training programmes that included short-term OR [50,52.55]. An increase in mCSA of the vastus lateralis was observed in male participants administered either whey protein with HMB-FA or with leucine during a 12-week RT programme comprising a 2-week OR phase (Jakubowski et al., 2019). No differences in mCSA were reported between groups at any phase, suggesting neither supplement was superior in invoking RT-induced muscle hypertrophy, and that OR itself had an impact on mCSA during periodized strength training. Combined HMB-FA plus ATP supplementation led to increased LBM (8.5 ± 0.8 kg vs. 2.1 ± 0.5 kg; $p < 0.05$) and thickness of the quadriceps muscle (7.8 ± 0.4 mm vs. 2.4 ± 0.7; $p < 0.001$) when compared to placebo during a 12-week strength training programme with 2-week OR (Lowery et al., 2016). Whilst this may appear to illustrate short-term

OR could have a positive impact on muscle hypertrophy, is worth noting that an increase in LBM of this magnitude in trained participants across 12-weeks of RT appears to be somewhat questionable and the authors of this review suggest these particular results be taken with caution. In another study, ATP supplementation was reported to enhance LBM (4.0 ± 0.4 kg vs, 2.1 ± 0.5 kg; $p < 0.009$) and quadriceps muscle thickness (4.9 ± 1.0 mm vs. 2.5 ± 0.6; $p < 0.02$) compared to placebo during a 12-week periodized strength training programme consisting of three phases of varied intensity and volume (Wilson et al., 2013) Concerns have since been raised regarding the methodological robustness of the data presented in the studies by Lowery et al., and Wilson et al., [52;55], with inconsistencies in data suggested by other researchers (Gentles & Phillips, 2017). To date, there is little evidence suggesting that purposeful OR during RT results in negative alterations in mCSA. However, this is likely due to a significant lack of literature looking specifically at markers of muscle hypertrophy during periods of RT resulting in NFOR or OTS. This provides an interesting gap for future research.

5. Conclusions

Implementing short-term OR is a common practice in some strength sports where the training stimulus is presented typically through high-volume/high-intensity RT over a 2–4 week period and often leads to FOR due to improved performance.

There is a lack of representation in the research from strength sports such as powerlifting, strongman and explosive throwing and sprinting sports, with only a single study assessing the effects of purposeful OR in throwing athletes.

Several performances, neuroendocrine, neuromuscular, and biochemical markers have been proposed as markers to determine NFOR/OTS in both strength sports and RT, but no single test or method has been able to identify the exact point at which FOR becomes NFOR or OTS. A dose–response transition from FOR to NFOR that is identifiable through perturbations in physiological markers or performance testing might exist; however, this has not been identified in the current literature.

There is no evidence that planned FOR protocols used in strength-sport research have led to true OTS based on current definitions (Meeusen et al., 2013). Studies have demonstrated short-term performance loss indicative of NFOR; however, in these cases, diminished performance has typically been resolved within days to weeks and is therefore not indicative of OTS. Many studies located for this review were published prior to the latest guidelines and definitions proposed by the ACSM/ECSS joint statement (Meeusen et al., 2013) and it appears that the definition of "overtraining" is poorly interpreted in the research. Reductions in performance lasting weeks to months, generalised performance reductions, and short-term high-volume/high-intensity training protocols have all been described as "overtraining", adding to the confusion around the true meaning of the term. That said, OTS may have occurred during prolonged exposure to high-intensity or combined high-intensity/high-volume RT.

Evidence suggests that NFOR is a real consequence of excessive and chronic training in the absence of sufficient recovery from RT, especially in practices involving extreme conditioning. Coaches and athletes must be cognisant to the deleterious effects of excessive training loads, and there is sufficient justification that a robust testing battery to identify when FOR becomes NFOR is now needed.

Specific areas of interest for further investigation include the effects of muscular failure training, weight cutting during periods of OR (which is a tool often used in weight category strength sports as well as bodybuilding) and the reliability of performance-specific measures and biomarker sensitivity to RT OTS.

Acknowledgments

The authors would like to thank Deborah Harrop and Dave Hembrough for their advice on the methodology and input into the designing and implementation of the scoping review.

Disclosure statement

No potential conflict of interest was reported by the authors.

ORCID

Lee Bell http://orcid.org/0000-0003-0583-3522
Alan Ruddock http://orcid.org/0000-0002-7001-9845
Thomas Maden-Wilkinson http://orcid.org/0000-0002-6191-045X
David. Rogerson http://orcid.org/0000-0002-4799-9865

References

Arksey, H., & O'Malley, L. (2005). Scoping studies: Towards a methodological framework. *International Journal of Social Research Methodology*, *8*(1), 19–32. https://doi.org/10.1080/1364557032000119616

Bazyler, C., Mizuguchi, S., Zourdos, M., Sato, K., Kavanaugh, A., DeWeese, B., Breuel, K. F., & Stone, M. H. (2017). Characteristics of a national level female weightlifter peaking for competition: A case study. *Journal of Strength and Conditioning Research*, *32*(11), 3029–3038. https://doi.org/10.1519/JSC.0000000000002379

Bazyler, C. D., Mizuguchi, S., Harrison, A. P., Sato, K., Kavanaugh, A. A., DeWeese, B. H., & Stone, M. H. (2017). changes in muscle architecture, explosive ability, and track and field throwing performance throughout a competitive season and after a taper. *Journal of Strength and Conditioning Research*, *31*(10), 2785–2793. https://doi.org/10.1519/JSC.0000000000001619

Berning, J. M., Adams, K. J., Climstein, M., & Stamford, B. A. (2007). Metabolic demands of "junkyard" training: Pushing and pulling a motor vehicle. *Journal Of Strength And Conditioning Research / National Strength & Conditioning Association*, *21*(3), 853.

Cadegiani, F. A., & Kater, C. E. (2017). Hormonal aspects of overtraining syndrome: A systematic review. *BMC Sports Science, Medicine and Rehabilitation*, *9*(1), 14–15. https://doi.org/10.1186/s13102-017-0079-8

Cadegiani, F. A., Kater, C. E., & Gazola, M. (2019). Clinical and biochemical characteristics of high-intensity functional training (HIFT) and overtraining syndrome: Findings from the EROS study (The EROS-HIFT). *Journal of Sports Sciences*, *37*(11), 1296–1307. https://doi.org/10.1080/02640414.2018.1555912

Davies, T., Orr, R., Halaki, M., & Hackett, D. (2016). Effect of training leading to repetition failure on muscular strength: A systematic review and meta-analysis. *Sports Medicine*, *46*(4), 487–502. https://doi.org/10.1007/s40279-015-0451-3

Drake, N., Smeed, J., Carper, M. J., & Crawford, D. A. (2017). Effects of short-term CrossFit[TM] training: A magnitude-based approach. *Journal Of Exercise Physiology Online / American Society of Exercise Physiologists*, *20*(2), 111. https://www.semanticscholar.org/paper/Effects-of-Short-Term-CrossFit-Training%3A-A-Approach-Baker-Brock/fab2fade89ce46d1da2a97faa1579c5d995dee38

Fahey, T. D., & Pearl, M. S. (1998). The hormonal and perceptive effects of phosphatidylserine administration during two weeks of resistive exercise-induced overtraining. *Biology Of Sport / Institute of Sport*, *15*(3), 135–144. https://www.bisp-surf.de/Record/PU199902307038/RIS

Fatouros, I. G., Destouni, A., Margonis, K., Jamurtas, A. Z., Vrettou, C., Kouretas, D., Mastorakos, G., Mitrakou, A., Taxildaris, K., Kanavakis, E., & Papassotiriou, I. (2006). Cell-free plasma DNA as a novel marker of aseptic inflammation severity related to exercise overtraining. *Clinical Chemistry*, *52*(9), 1820–1824. https://doi.org/10.1373/clinchem.2006.070417

Fry, A. C., & Kraemer, W. J. (1997). Resistance exercise overtraining and overreaching. *Sports Medicine*, *23*(2), 106–129. https://doi.org/10.2165/00007256-199723020-00004

Fry, A. C., Kraemer, W. J., Lynch, J. M., & Barnes, J. M. (2001). Overload injury of the knees with resistance-exercise overtraining: A case study. *Journal of Sport Rehabilitation*, *10*(1), 57–66. https://doi.org/10.1123/jsr.10.1.57

Fry, A. C., Kraemer, W. J., Lynch, J. M., Triplett, N. T., & Koziris, L. P. (1994c). Does short-term near-maximal intensity machine resistance training induce overtraining? *Journal Of Strength And Conditioning Research / National Strength & Conditioning Association*, *8*(3), 188. https://journals.lww.com/nsca-jscr/abstract/1994/08000/does_short_term_near_maximal_intensity_machine.11.aspx

Fry, A. C., Kraemer, W. J., & Ramsey, L. T. (1998). Pituitary-adrenal-gonadal responses to high-intensity resistance exercise overtraining. *Journal of Applied Physiology*, *85*(6), 2352–2359. https://doi.org/10.1152/jappl.1998.85.6.2352

Fry, A. C., Kraemer, W. J., Stone, M. H., Koziris, L. P., Thrush, J. T., & Fleck, S. J. (2000a). Relationships between serum testosterone, cortisol, and weightlifting performance. *The Journal of Strength and Conditioning Research*, *14*(3), 338–343. https://doi.org/10.1519/1533-4287(2000)014<0338:RBSTCA>2.0.CO;2

Fry, A. C., Kraemer, W. J., Stone, M. H., Warren, B. J., Fleck, S. J., Kearney, J. T., & Gordon, S.E. (1994a). Endocrine responses to overreaching before and after 1 year of weightlifting, *Can J Appl Physiol, 19*(4), 400–410. https://www.nrcresearchpress.com/doi/10.1139/h94-032#.XrK5E2jdvIU

Fry, A. C., Kraemer, W. J., Stone, M. H., Warren, B. J., Kearney, J. T., Maresh, C. M., Weseman, C. A., & Fleck, S. J. (1993). Endocrine and performance responses to high volume training and amino acid supplementation in elite junior weightlifters. *International Journal of Sport Nutrition*, *3*(3), 306–322. https://doi.org/10.1123/ijsn.3.3.306

Fry, A. C., Kraemer, W. J., Van Borselen, F., Lynch, J. M., Marsit, J. L., Roy, E. P., Triplett, N. T., & Knuttgen, H. G. (1994b). Performance decrements with high-intensity resistance exercise overtraining. *Medicine & Science in Sports & Exercise*, *26*(9), 1165–1173. https://doi.org/10.1249/00005768-199409000-00015

Fry, A. C., Kraemer, W. J., Van Borselen, F., . J., Lynch, J. M., Triplett, N. T., Koziris, L. P., & Fleck, S. J. (1994d). Catecholamine responses to short-term high-intensity resistance exercise overtraining. *Journal of Applied Physiology*, *77*(2), 941–946. https://doi.org/10.1152/jappl.1994.77.2.941

Fry, A. C., Schilling, B. K., Weiss, L. W., & Chiu, L. (2006). β2-Adrenergic receptor downregulation and performance decrements during high-intensity resistance exercise overtraining. *Journal of Applied Physiology*, *101*(6), 1664–1672. https://doi.org/10.1152/japplphysiol.01599.2005

Fry, A. C., Webber, J. M., Weiss, L. W., Fry, M. D., & Li, Y. (2000b). Impaired performances with excessive high-intensity free-weight training. *Journal Of Strength And Conditioning Research / National Strength & Conditioning Association*, *14*(1), 54–61. https://journals.lww.com/nsca-jscr/Abstract/2000/02000/Impaired_Performances_with_Excessive.10.aspx

Gentles, J. A., & Phillips, S. M. (2017). Discrepancies in publications related to HMB-FA and ATP supplementation. *Nutrition & Metabolism*, *14*(1), 42. https://doi.org/10.1186/s12986-017-0201-7

Haff, G., Jackson, J., Kawamori, N., Carlock, J., Hartman, M., Kilgore, J., Morris, R. T., Ramsey, M. W., Sands, W. A., & Stone, M. H. (2008, March). Force-time curve characteristics and hormonal alterations during an eleven-week training period in elite women weightlifters. *Journal of Strength and Conditioning Research*, *22*(2), 433–446. https://doi.org/10.1519/JSC.0b013e31816191be

Häkkinen, K., Keskinen, K. L., Alén, M., Komi, P. V., & Kauhanen, H. (1989). Serum hormone concentrations during prolonged training in elite endurance-trained and strength-trained athletes. *European Journal of Applied Physiology and Occupational Physiology*, *59*(3), 233–238. https://doi.org/10.1007/BF02386193

Häkkinen, K., Pakarinen, A., Alén, M., Kauhanen, H., & Komi, P. V. (1987). Relationships between training volume, physical performance capacity, and serum hormone concentrations during prolonged training in elite weight lifters. *International Journal of Sports Medicine*, *8*(S 1), S6–S65. https://doi.org/10.1055/s-2008-1025705

Halson, S. L., & Jeukendrup, A. E. (2004). Does overtraining exist?: An analysis of overreaching and overtraining research. *Sports Medicine*, *34*(14), 967–981. https://doi.org/10.2165/00007256-200434140-00003

Hartman, M. J., Clark, B., Bembens, D. A., Kilgore, J. L., & Bemben, M. G. (2007). Comparisons between twice-daily and once-daily training sessions in male weight lifters. *International Journal of Sports Physiology and Performance*, *2*(2), 159–169. https://doi.org/10.1123/ijspp.2.2.159

Jakubowski, J., Wong, E. P., Nunes, E., Noguchi, K., Vandeweerd, J., Murphy, K., Morton, R. W., Mcglory, C., & Phillips, S. M. (2019). Equivalent hypertrophy and strength gains in β-hydroxy-β-methylbutyrate- or leucine-supplemented men. *Medicine & Science in Sports & Exercise*, *51*(1), 65–74. https://doi.org/10.1249/MSS.0000000000001752

Khlif, R., Marrakchi, R., Jamoussi, K., Sahnoun, Z., Chtourou, H., & Souissi, N. (2019). Plasma iron status in elite weightlifters after four weeks of intensive training. *Science & Sports*, *34*(5), 328.e1-328.e8. https://doi.org/10.1016/j.scispo.2019.02.003

Kraemer, W. J., & Ratamess, N. A. (2005). Hormonal responses and adaptations to resistance exercise and training. *Sports Medicine*, *35*(4), 339–361. https://doi.org/10.2165/00007256-200535040-00004

Kraemer, W. J., Ratamess, N. A., Volek, J. S., Häkkinen, K., Rubin, M. R., French, D. N., Gómez, A. L., McGuigan, M. R., Scheett, T. P., Newton, R. U., Spiering, B. A., Izquierdo, M., & Dioguardi, F. S. (2006). The effects of amino acid supplementation on hormonal responses to resistance training overreaching. *Metabolism*, *55*(3), 282–291. https://doi.org/10.1016/j.metabol.2005.08.023

Kreher, J. B. (2016). Diagnosis and prevention of overtraining syndrome: An opinion on education strategies. *Open Access Journal of Sports Medicine*, *7*, 115–122. https://doi.org/10.2147/OAJSM.S91657

Levac, D., Colquhoun, H., & O'Brien, K. K. (2010). Scoping studies: Advancing the methodology. *Implementation Science*, *5*(1), 69. https://doi.org/10.1186/1748-5908-5-69

Lowery, R., Joy, J., Rathmacher, J., Baier, S., Fuller, J., John, S., Mack, I., Purpura, M., Wilson, S. M. C., & Wilson, J. M. (2016). Interaction of beta-hydroxy-beta-methylbutyrate free acid and adenosine triphosphate on muscle mass, strength, and power in resistance trained individuals. *Journal of Strength and Conditioning Research*, *30*(7), 1843–1854. https://doi.org/10.1519/JSC.0000000000000482

Meeusen, R., Duclos, M., Foster, C., Fry, A., Gleeson, M., Nieman, D., Raglin, J., Rietjens, G., Steinacker, J., & Urhausen, A. (2013). Prevention, diagnosis and treatment of the overtraining syndrome: Joint consensus statement of the european college of sport science (ECSS) and the American college of sports medicine (ACSM). *European Journal of Sport Science*, *13*(1), 1–24. https://doi.org/10.1080/17461391.2012.730061

Morán-Navarro, R., Pérez, C., Mora-Rodríguez, R., de la Cruz-sánchez, E., González-Badillo, J., Sánchez-Medina, L., & Pallarés, J. G. (2017). Time course of recovery following resistance training leading or not to failure. *European Journal of Applied Physiology*, *117*(12), 2387–2399. https://doi.org/10.1007/s00421-017-3725-7

Munn, Z., Peters, M. D. J., Stern, C., Tufanaru, C., McArthur, A., & Aromataris, E. (2018). Systematic review or scoping review? Guidance for authors when choosing between a systematic or scoping review approach. *BMC Medical Research Methodology*, *18*(1), 1–7. https://doi.org/10.1186/s12874-018-0611-x

Nicoll, J., Fry, A., Galpin, A., Sterczala, A., Thomason, D., Moore, C., Weiss, L. W., & Chiu, L. Z. F. (2016). Changes in resting mitogen-activated protein kinases following resistance exercise overreaching and overtraining. *European Journal of Applied Physiology*, *116*(11), 2401–2413. https://doi.org/10.1007/s00421-016-3492-x

Peters, M. D. J., Godfrey, C., McInerney, P., Baldini Soares, C., Khalil, H., & Parker, D. (2017). Chapter 11: Scoping Reviews. In E. Aromataris & Z. Munn (Eds.), *Joanna briggs institute reviewer's manual*. The Joanna Briggs Institute.

Peters, M. D. J., Godfrey, C. M., Khalil, H., McInerney, P., Parker, D., & Soares, C. B. (2015). Guidance for conducting systematic scoping reviews. *International Journal of Evidence-based Healthcare*, *13*(3), 141–146. https://doi.org/10.1097/XEB.0000000000000050

Pistilli, E. E., Kaminsky, D. E., Totten, L. M., & Miller, D. R. (2008). Incorporating one week of planned overreaching into the training program of weightlifters. *Strength and Conditioning Journal*, *30*(6), 39–44. https://doi.org/10.1519/SSC.0b013e31818ee78c

Raastad, T., Glomsheller, T., Bjøro, T., & Hallén, J. (2001). Changes in human skeletal muscle contractility and hormone status during 2 weeks of heavy strength training. *European Journal of Applied Physiology*, *84*(1), 54–63. https://doi.org/10.1007/s004210000328

Raastad, T., & Hallén, J. (2000). Recovery of skeletal muscle contractility after high- and moderate-intensity strength exercise. *European Journal of Applied Physiology*, *82*(3), 206–214. https://doi.org/10.1007/s004210050673

Raeder, C., Wiewelhove, T., Simola, R. Á. D., Kellmann, M., Meyer, T., Pfeiffer, M., & Ferrauti, A. (2016). Assessment of fatigue and recovery in male and female athletes after 6 days of intensified strength training. *Journal of Strength and Conditioning Research*, *30*(12), 3412–3427. https://doi.org/10.1519/JSC.0000000000001427

Ratamess, N. A., Kraemer, W. J., Volek, J. S., Rubin, M. R., Gómez, A. L., French, D. N., Sharman, M. J., McGuigan, M. M., Scheett, T., Häkkinen, K., Newton, R. U., & Dioguardi, F. (2003). The effects of amino acid supplementation on muscular performance during resistance training overreaching. *Journal of Strength and Conditioning Research*, *17*(2), 250–258. https://doi.org/10.1519/1533-4287(2003)017<0250:teoaas>2.0.co;2

Robbins, D., Marshall, P., & McEwen, M. (2012). The effect of training volume on lower-body strength. *Journal of Strength and Conditioning Research*, *26*(1), 34–39. https://doi.org/10.1519/JSC.0b013e31821d5cc4

Sharp, C., & Pearson, D. (2010). Amino acid supplements and recovery from high-intensity resistance training. *Journal of Strength and Conditioning Research*, *24*(4), 1125–1130. https://doi.org/10.1519/JSC.0b013e3181c7c655

Sikorski, E., Wilson, J., Lowery, R., Joy, J., Laurent, C., Wilson, S., Hesson, D., Naimo, M. A., Averbuch, B., & Gilchrist, P. (2013). Changes in perceived recovery status scale following high-volume muscle damaging resistance exercise. *Journal of Strength and Conditioning Research*, *27*(8), 2079–2085. https://doi.org/10.1519/JSC.0b013e31827e8e78

Soares, S., Ferreira-Junior, J., Pereira, M., Cleto, V., Castanheira, R., Cadore, E., Brown, L. E., Gentil, P., Bemben, M. G., & Bottaro, M. (2015). Dissociated time course of muscle damage recovery between single- and multi-joint exercises in highly resistance-trained men. *Journal of Strength and Conditioning Research*, *29*(9), 2594–2599. https://doi.org/10.1519/JSC.0000000000000899

Sterczala, A., Fry, A., Chiu, L., Schilling, B., Weiss, L., & Nicoll, J. (2017). ß-Adrenergic receptor maladaptations to high power resistance exercise overreaching. *Human Physiology*, *43*(4), 446–454. https://doi.org/10.1134/S0362119717040144

Stone, M. H., Pierce, E., Kyle, C., Sands, W. A., & Stone, M. E. (2006). Weightlifting: Program design. *Strength and Conditioning Journal*, *28*(2), 10–17. https://journals.lww.com/nsca-scj/Abstract/2006/04000/Weightlifting__Program_Design.1.aspx

Stone, M. H., Potteiger, J., Pierce, K., Proulx, C., O'Bryant, H. S., Johnson, R. L., & Stone, M. E. (2000). Comparison of the effects of three different weight-training programs on the one repetition maximum squat. *Journal Of Strength And Conditioning Research / National Strength & Conditioning Association*, *14*(3), 332–337. https://insights.ovid.com/strength-conditioning-research/jscr/2000/08/000/comparison-effects-three-different-weight-training/15/00124278

Suarez, D. G., Mizuguchi, S., Hornsby, W. G., Cunanan, A. J., Marsh, D. J., & Stone, M. H. (2019). Phase-specific changes in rate of force development and muscle morphology throughout a block periodized training cycle in weightlifters. *Sports (Basel, Switzerland)*, *7*(6), 129. https://www.mdpi.com/2075-4663/7/6/129

Szewczyk, A., Rebowska, E., & Jegier, A. (2018). Prevalence of non-functional overreaching and overtraining syndromes in athletes. *Polish Journal of Sports Medicine*, *34*(4), 213–218. https://medycynasportowa.edu.pl/resources/html/article/details?id=184175&language=pl

Taylor, K., Hopkins, W. G., Chapman, D. W., & Cronin, J. B. (2016). The influence of training phase on error of measurement in jump performance. *International Journal of Sports Physiology and Performance*, *11*(2), 235–239. https://doi.org/10.1123/ijspp.2015-0115

Tibana, R. A., de Almeida, L. M., Frade de Sousa, N. M., Nascimento, D., Neto, I. V., de Almeida, J. A., de Souza, V. C., Lopes, M. D. F. T. P. L., Nobrega, O. D. T., Vieira, D. C. L., Navalta, J. W., & Prestes, J. (2018). Corrigendum: Two consecutive days of extreme conditioning program training affects pro and anti-inflammatory cytokines and osteoprotegerin without impairments in muscle power. *Frontiers in Physiology*, *9*, 771. https://doi.org/10.3389/fphys.2018.00771

Todd, G., Taylor, J. L., & Gandevia, S. C. (2003). Measurement of voluntary activation of fresh and fatigued human muscles using transcranial magnetic stimulation. *The Journal of Physiology*, *551*(2), 661–671. https://doi.org/10.1113/jphysiol.2003.044099

Tricco, A. C., Lillie, E., Zarin, W., O'Brien, K. K., Colquhoun, H., Levac, D., Moher, D., Peters, M. D. J., Horsley, T., Weeks, L., Hempel, S., Akl, E. A., Chang, C., McGowan, J., Stewart, L., Hartling, L., Aldcroft, A., Wilson, M. G., Garritty, C., Lewin, S., & Straus, S. E. (2018). PRISMA extension for scoping reviews (PRISMA-ScR): Checklist and explanation. *Annals of Internal Medicine*, *169*(7), 467. https://doi.org/10.7326/M18-0850

Triscott, S., Gordon, J., Kuppuswamy, A., King, N., Davey, N., & Ellaway, P. (2008). Differential effects of endurance and resistance training on central fatigue. *Journal of Sports Sciences*, *26*(9), 941–951. https://doi.org/10.1080/02640410701885439

Volek, J., Ratamess, N., Rubin, M., Gómez, A., French, D., McGuigan, M., Scheett, T. P., Sharman, M. J., Hkkinen, K., & Kraemer, W. J. (2004). The effects of creatine supplementation on muscular performance and body composition responses to short-term resistance training overreaching. *European Journal of Applied Physiology*, *91*(5), 628–637. https://doi.org/10.1007/s00421-003-1031-z

Warren, B. J., Stone, M. H., Kearney, J. T., Fleck, S. J., Johnson, R. L., Wilson, G. D., & Kraemer, W. (1992). Performance measures, blood lactate and plasma ammonia as indicators of overwork in elite junior weightlifters. *International Journal of Sports Medicine*, *13*(5), 372–376. https://doi.org/10.1055/s-2007-1021283

Willardson, J. M. (2007). The application of training to failure in periodized multiple-set resistance exercise programs. *Journal Of Strength And Conditioning Research / National Strength & Conditioning Association*, *21*(2), 628. https://europepmc.org/article/med/17530977

Willardson, J. M., Norton, L., & Wilson, G. (2010). Training to failure and beyond in mainstream resistance exercise programs. *Strength and Conditioning Journal*, *32*(3), 21–29. https://doi.org/10.1519/SSC.0b013e3181cc2a3a

Wilson, J. M., Joy, J. M., Lowery, R. P., Roberts, M. D., Lockwood, C. M., Manninen, A. H., Fuller, J. C., De Souza, E. O., Baier, S. M., Wilson, S. M., & Rathmacher, J. A. (2013). Effects of oral adenosine-5′-triphosphate supplementation on athletic performance, skeletal muscle hypertrophy and recovery in resistance-trained men. *Nutrition and Metabolism*, *10*(1), 57. https://doi.org/10.1186/1743-7075-10-57